Inside an Investment Bank

Politecnico di Milano

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01

Organization

Which roles you can find in a bank

An investment bank is not one monolithic entity. It is a collection of distinct but interconnected business divisions and functional layers, covering several aspects of the financial world.

Business Divisions

IBD

M&A advisory, Capital raising.
It structures strategic transactions as leveraged buyouts, equity transactions (IPOs) and debt offering (bond issuances).

Markets

Sales & Trading, Structuring derivatives.
Develop institutional and corporate client relationship and cover product ideation, recommendation and execution.

Asset & Wealth Management

Portfolio management, Client advisory.
Deliver strategies with disciplined risk management (funds, sovereigns) and tailored solutions (HNW).

Transaction Services

Treasury, Cash management, Trade finance.
Covers pyment, cash, liquidity, treasury solutions and cross border trade operations (e.g. delivery of commodity).

Support functions

Finance & Legal, Risk & Compliance, Technology & Operations, Research& Planning.
Firm-wide functions to support its control, grow and resilience.

Three functional layers

Regardless of the business division, every function typically falls into one of three layers.

Front office

Level that faces clients and provides them services, such as market operations, advisory, investment recommendation.
It's the bank revenues engine, originating business and executing deals.

Middle office

Supports FO providing everything need to function.
Monitor all type of financial risk, market, counterparty, legal, operational, liquidity...
Ensure law and regulatory compliance.
Validate trades and ensure correctness.

Back office

The operational backbone—trade processing, settlements, accounting, technology, data infrastructure and core infrastructure support.
Everything needed by the bank to be operative.

02

Market Unit

Sales & Trading, Structuring

Work with clients, multinational corporates, small and medium enterprises, institutional and private. Offer OTC derivatives (plain vanilla and structured) and investment products (securities, funds and structured notes) that fulfill the client risk management, financing and investment needs. Make money on fees, commissions and bid ask spread.

Usually focus on a single asset class:

- Equity (cash, derivatives, delta one...)
- Credit (corporate bonds, loans, CDS...)
- Bonds (government and SSA)
- Flow Rates (IRS and CCS, G10 currencies, other currencies...)
- Structured rates (plain vanilla options, more exotic structures...)
- FX (G10, EMEA, Asia, LatAm...)
- FX option (options, TARF, barrier, exotic Fx structures...)
- Commodity (power, oil, gas, metal, agriculture, EUA...)

1 **Sales** — Sales maintain the relationship with the client, pitch products and trading ideas that might meet its needs.

2 **Structuring** — Structuring create tailored solutions for the client whenthe standard product are not flexible enough.

3 **Trading** — Trading act on the market on behalf of the client (flow) or take positions in the market to make profit (prop).

03

Risk, Technology & Research

Quants and Risk

Front office / Desk quants

FO Quants work in synergy with Sales & Trading, developing real time pricing models, execution algorithms, trading strategies. Their work directly impacts revenue, and the pace is fast.

Middle office / Back office quants

Their responsibilities include model and PnL validation, regulatory capital modeling, risk reporting, data integrity management.
The timelines are longer and the focus is on rigor, governance, in order to be compliant with regulatory requirements.

Risk unit

Risk unit cover all the type of risks, in compliance with the regulatory framework: market, credit, operational, liquidity, legal, environmental.
It creates, develops and validates models.
It deals with banking authority revisions.
It is responsible of VaR, stress test, backtesting.
Risk supervises trading desks behavior, validating their pnl, approving limits on their sensitivities and reporting breaches.

The Black Box Revealed

DATA

Prices, volumes, characteristics, time series, unstructured data

Powers the entire black box machinery

Alpha Model

Predicts the future returns of the instruments the quant wants to consider trading.

Risk Model

Limits exposure to factors unlikely to generate returns but that could drive losses.

Transaction Cost Model

Estimates cost of migrating to target portfolio. Costs take different forms: bid/ask spread, slippage, and market impact.

Portfolio Construction

Balances profits, risk limits, and transaction costs to determine the optimal portfolio

Execution

Low-cost, efficient trade implementation

Asset Class Return Drivers

Equities

$$R \approx \text{Dividend Yield} + \text{Dividend Growth} + \text{Valuation Change}$$

Hist: ~11%

Equity premium over cash: ~7.5%

~2% Div. Yield (today)

Dividends relative to price paid. Historically averaged 3.9% (U.S. 1926–2013). Includes net share buybacks.

~3.5% Div. Growth (nominal)

Driven by inflation (~2%) plus real economic growth (~1.5%). Historically 4.6% due to higher past inflation.

2.4% Valuation Change (hist.)

Changes in price-dividend ratio. Cannot grow forever. Mean-reversion likely reduces future contribution.

Bonds

$$R \approx YTM - Dur \times \Delta YTM$$

Long-run return equals the yield to maturity. Short-run returns depend on yield changes, driven by surprises in growth, inflation, or monetary policy. The roll-down effect assumes an unchanged yield curve.

Credit

$$R \approx Spread - Dur \times \Delta S - Loss$$

Credit return equals the spread, minus duration-weighted spread changes, minus expected default losses. A higher credit spread leads to a higher return, everything else being equal. Spread narrowing produces capital appreciation; widening causes losses.

Currency

$$E(R) \approx r^* - r - \Delta\pi + PPP$$

Short-run return equals carry (foreign interest rate) plus currency appreciation. The carry trade exploits the failure of uncovered interest rate parity. Long-run return depends on the real interest rate differential and convergence to purchasing power parity.